

A simpler construction of square-difference-free sets beyond exponent 0.758

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Abstract

We construct subsets of $\{1, \dots, N\}$ with no nonzero square difference and cardinality at least $N^{0.758001-o(1)}$. The construction improves on Krachun's exponent 0.752796455874514... and is a simpler companion to our construction of exponent 0.75806746. Its purpose is to make the mechanism accessible: allow modular square differences, order them with intervals, and combine the resulting local objects by the Chinese remainder theorem. A short stopping-tree argument turns interval moments into an exponent. The concrete ingredients are six small prime chains, two odd components with only 17 and 729 low points, and an explicit seven-state binary recursion. All general implications are proved here. The remaining finite checks use exact rational arithmetic; their complete data and a standalone verifier accompany this paper in its source and PDF attachment.

1 Introduction

For $N \geq 1$, let

$$D(N) = \max\{|A| : A \subseteq \{1, \dots, N\}, (A - A) \cap \{1, 4, 9, \dots\} = \emptyset\}.$$

The Furstenberg–Sárközy theorem says that $D(N) = o(N)$ [1, 2]. We study the complementary question: how large an exponent α can a construction achieve in $D(N) \geq N^{\alpha-o(1)}$? This notation means that for every $\varepsilon > 0$ the inequality $D(N) \geq N^{\alpha-\varepsilon}$ holds for every sufficiently large integer N .

Theorem 1.1. *With $\alpha_0 = 758001/1000000 = 0.758001$, we have*

$$D(N) \geq N^{\alpha_0-o(1)}.$$

Previous constructions. Ruzsa obtained the exponent

$$\frac{1}{2} + \frac{\log 7}{2 \log 65} = 0.733077 \dots$$

by an alternating-digit construction [3]. Beigel and Gasarch, and independently Lewko, obtained

$$\frac{1}{2} + \frac{\log 12}{2 \log 205} = 0.7334117970 \dots$$

using a 12-element square-difference-free set modulo 205 [5, 6]. The often-mentioned three-quarter barrier concerns this particular modular-avoidance strategy and conjectured limitations on its inputs; it is not an upper bound on $D(N)$.

Krachun’s recent construction crosses that barrier [9]. Instead of requiring local residue sets to avoid all square differences, it permits such differences when they consistently increase a rank (in our sign convention). Ranks from coprime components add, while the numbers of digits multiply. His explicit prime-chain inputs give

$$\alpha_K = \frac{10 + \sum_{i=1}^{10} \log p_i / \log t_i}{1 + 2 \sum_{i=1}^{10} \log p_i / \log t_i} = 0.752796455874514 \dots, \quad (1)$$

where

$$(p_i, t_i) = (3, 2), (7, 3), (11, 4), (19, 5), (23, 5), (31, 7), (43, 7), (59, 9), (71, 9), (103, 11).$$

Our theorem raises this exponent by approximately 0.005204544125486. The ranked realization, repetition, and CRT mechanism are due to Krachun; we prove the needed statements below, with the reversed rank convention [9, Lemmas 4–5].

For perspective, the upper-bound problem has developed through the work of Pintz, Steiger and Szemerédi, Bloom and Maynard, and Green and Sawhney [4, 7, 8]. Green and Sawhney prove $D(N) \ll N \exp(-c\sqrt{\log N})$ for an absolute $c > 0$ [8, Theorem 1.1]. There remains a large gap between these upper bounds and the construction problem considered here.

The extra flexibility. Replace a discrete rank by an interval. If $y - x$ is a square modulo the local modulus, require the interval of x to lie to the left of that of y . Intervals for incomparable digits may overlap, and their lengths need not agree. Repeating digits multiplies interval lengths; this makes *moments of the lengths* the natural measure of a local construction. For a target exponent α , a component in a square base b earns a contribution f when its widths satisfy

$$\sum_x w_x^f \geq b^\alpha.$$

Section 3 proves that coprime components achieve the target as soon as their contributions add to more than α . The proof uses a finite tree whose mass is conserved at every branching. No limiting frequency argument or optimization theorem is required.

The two additional local ideas are to group odd primes and to use the prime 2. For odd primes, restricted digits are interleaved with free digits; the first differing position is inspected separately at each prime. For 2, a state records where the even and odd digit intervals may lie. The fact that odd squares are 1 modulo 8 then leads to a small recursion. The moment method accommodates both kinds of inputs in the same argument.

Why this companion is simpler. A separate manuscript attains 0.75806746 with larger finite witnesses [12]. Here the target is deliberately lower: the two odd supports have 17 and 729 points, instead of 4,913 and 19,683, and the binary recursion has seven states and 26 branches, instead of 25 states and 94 branches. Its windows, transitions, and positive comparison vector fit in explicit tables. The final budget is the exact addition

$$0.51011 + 0.0268 + 0.0673 + 0.154 = 0.75821 > 0.758001. \quad (2)$$

This paper proves its own result without using the stronger companion.

The theorem remains computer-assisted: the 729-point component carries a finite table of widths. We specify how those data define the construction, prove why the verification conditions suffice, and supply the complete checks. Optimization helped find the inputs; no optimizer or floating-point comparison is needed to accept them. All logarithms are natural, and all terminating decimals specifying parameters denote exact rationals.

2 From local order to square-difference-free sets

The local goal is an order compatible with square differences, rather than an immediate square-difference-free set. The final lift will put a higher rank into an earlier integer block, turning every potential positive square difference into a negative number.

For $b \geq 1$, let $\mathcal{Q}(b) = \{z^2 \bmod b : z \in \mathbb{Z}\}$, including zero. Residues are represented canonically in $\{0, \dots, b-1\}$. For distinct x, y write $x \rightarrow_b y$ if $y - x \in \mathcal{Q}(b)$. For composite b , a nonzero square residue may vanish in some prime coordinates; invertibility is never imposed.

Definition 2.1. A ranked alphabet modulo b is a finite set C of canonical residues with an integer function $r : C \rightarrow \{0, \dots, h-1\}$, for some $h \geq 1$, such that $x \rightarrow_b y$ implies $r(x) < r(y)$. An ordered interval alphabet assigns to every $x \in C$ an interval $[a_x, a_x + w_x] \subseteq [0, 1]$, with $w_x > 0$, such that $x \rightarrow_b y$ implies $a_x + w_x \leq a_y$. Its moment of order $f \geq 0$ is $Z_f(C, w) = \sum_{x \in C} w_x^f$.

Intervals may overlap when the corresponding digits are incomparable. The graph of an ordered interval alphabet is acyclic, since a directed cycle would imply that the sum of its positive widths is at most zero.

The small abstract example in Figure 1 illustrates the flexibility. If the only required arrows are $a \rightarrow b \rightarrow c$ and $d \rightarrow c$, the interval for d can occupy two thirds of the unit interval. This is compatible with the shorter intervals for a and b . The same idea applies to the directed modular-square graphs used in our construction.

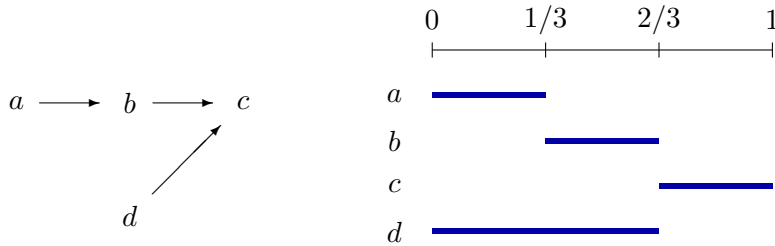


Figure 1: Intervals may overlap when no order is required. The moment here is $3(1/3)^f + (2/3)^f$. This is an illustrative directed graph, not a separate arithmetic input.

Lemma 2.2 (Path budgets). *Let a finite directed acyclic graph have positive vertex weights w_x . There are intervals in $[0, 1]$ of those widths ordered along every edge if and only if the sum of the weights on every directed path is at most one.*

Proof. Ordered intervals along a path have disjoint interiors, so their widths sum to at most one. Conversely, let d_x be the maximum weight of a path ending at x , including x . Set $a_x = d_x - w_x$. Then $a_x \geq 0$ and $a_x + w_x = d_x \leq 1$. If $x \rightarrow y$, appending y to a maximizing path ending at x gives $d_y \geq d_x + w_y$, whence $a_y \geq a_x + w_x$. $\square$

This is the weighted-chain viewpoint underlying the chain polytope of a finite poset [10]; the elementary proof above supplies the part used here.

Lemma 2.3 (Reverse-rank realization). *A ranked alphabet of size q modulo M , with ranks in $\{0, \dots, h-1\}$, gives a square-difference-free set of size q in $\{1, \dots, Mh\}$.*

Proof. For the canonical residue x , choose

$$T(x) = 1 + x + M(h - 1 - r(x)).$$

These numbers lie in the stated interval and are distinct modulo M . If $T(y) - T(x) = z^2 > 0$, then $x \neq y$ and reduction modulo M implies $r(y) - r(x) \geq 1$. Consequently

$$T(y) - T(x) = y - x - M(r(y) - r(x)) \leq (M - 1) - M < 0,$$

a contradiction. $\square$

Lemma 2.4 (CRT gluing). *Suppose $M_1, \dots, M_n$ are pairwise coprime and C_i is a ranked alphabet modulo M_i , with q_i digits and ranks below h_i . Their CRT product is a ranked alphabet modulo $M = \prod_i M_i$, with $Q = \prod_i q_i$ digits and ranks below*

$$h = 1 + \sum_{i=1}^n (h_i - 1).$$

Proof. The Chinese remainder map identifies residues modulo M with tuples of residues modulo the M_i . Explicitly, writing $P_i = M/M_i$, choose u_i with $P_i u_i \equiv 1 \pmod{M_i}$ and take the canonical residue of $\sum_i x_i P_i u_i$. This proves existence; divisibility by every M_i proves uniqueness. Assign rank $r(x) = \sum_i r_i(x_i)$. A square difference modulo M is a square in each coordinate. Equal coordinates leave the rank unchanged and unequal ones increase it. At least one coordinate is unequal, so the sum strictly increases. The size and rank bounds follow. $\square$

Lemmas 2.3 and 2.4 express the ranked realization and gluing mechanism of [9, Lemmas 4–5] in our convention.

Lemma 2.5 (Square-base words). *Let $b = s^2 > 1$. If digits C form a ranked alphabet with rank height h , then their length- k words form a ranked alphabet modulo b^k with $|C|^k$ digits and rank height h^k . If the digits instead carry ordered intervals, the word intervals obtained by affine composition are ordered along modular square differences as well.*

Proof. Encode a word by $X = \sum_{j=0}^{k-1} x_j b^j$. Uniqueness of the base- b expansion gives injectivity. Suppose $Y - X \equiv z^2 \pmod{b^k}$ and $X \neq Y$. Let j be the first position at which $x_j \neq y_j$, reading from the least significant digit. Since $s^{2j} = b^j$ divides z^2 , we have $s^j \mid z$: this follows by comparing the valuations of every prime dividing s . Dividing the congruence by b^j and reducing modulo b gives $y_j - x_j \equiv (z/s^j)^2 \pmod{b}$, a nonzero square edge.

In the ranked case assign the word rank $R(X) = \sum_{j=0}^{k-1} r(x_j) h^{k-1-j}$. The first differing digit increases its rank, and its place value dominates every possible decrease in the remaining places. Thus $R(Y) > R(X)$. Its range is $0 \leq R < h^k$.

For intervals, put $F_x(t) = a_x + w_x t$ and assign $F_{x_0} \circ \dots \circ F_{x_{k-1}}([0, 1])$. At the first differing position, the two digit intervals are ordered. The remaining compositions lie inside those digit intervals, and the common preceding maps are increasing. This preserves the required order. The width is $W_X = \prod_{j=0}^{k-1} w_{x_j}$. $\square$

Lemma 2.6 (Rounding intervals to ranks). *If every interval in an ordered alphabet has width at least $\gamma > 0$, then $r(x) = \lfloor a_x / \gamma \rfloor$ is a valid rank with $0 \leq r(x) < \lceil 1/\gamma \rceil$.*

Proof. On an edge $a_y \geq a_x + \gamma$, so $r(y) \geq r(x) + 1$. Moreover $0 \leq a_x \leq 1 - \gamma$, which gives the stated upper bound. $\square$

3 A finite interval-moment criterion

Theorem 3.1. *Let $n \geq 1$. For each $i \in \{1, \dots, n\}$ let C_i be an ordered interval alphabet in a square base $b_i > 1$, with the bases pairwise coprime. Suppose there are common constants $0 < \sigma \leq \rho < 1$ such that $\sigma \leq w_{i,x} \leq \rho$ for all digits. If $\alpha \geq 0$, $f_i \geq 0$, and*

$$Z_{f_i}(C_i, w_i) \geq b_i^\alpha \quad (1 \leq i \leq n), \quad F := \sum_{i=1}^n f_i > \alpha,$$

then $D(N) \geq N^{\alpha-o(1)}$.

The reason for the strict surplus is simple. At scale δ , the number of available words is controlled by $\delta^{-\sum f_i}$, whereas the common rank height costs only a factor of order δ^{-1} . The strict inequality $\sum f_i > \alpha$ pays for this height and for selecting a common length within each component.

The common width bounds are automatic for a fixed finite collection of nonempty alphabets with every width strictly between zero and one. They need not be uniform over a sequence of proposed constructions.

Lemma 3.2 (Stopping at a common width). *For one alphabet in base b , suppose $\sigma \leq w_x \leq \rho < 1$ and $Z = \sum_x w_x^f \geq q = b^\alpha$, with $f \geq 0$. For every integer $K \geq 1$, put $\delta = \rho^K$. There are a length $1 \leq k \leq K$ and a set D of length- k words such that*

$$\sigma\delta < W_X \leq \delta \quad (X \in D), \quad q^k \leq K|D|\delta^f.$$

Proof. Starting at the empty word, extend a word by every digit until its width first becomes at most δ . Every branch stops by depth K , since a length- K word has width at most ρ^K . A stopped word has a parent of width greater than δ , so its width is greater than $\sigma\delta$.

Give the edge labelled x the probability w_x^f/Z . These probabilities sum to one. The mass of a word X of length k is therefore W_X^f/Z^k . Replacing a node by all its children preserves its mass, so the terminal masses in this finite tree sum to one. These are masses on words; overlap between their geometric intervals causes no difficulty. Let T_k be the sum of W_X^f over the words stopping at depth k . Because $Z \geq q$,

$$1 = \sum_{k=1}^K \frac{T_k}{Z^k} \leq \sum_{k=1}^K \frac{T_k}{q^k}.$$

At least one summand on the right is at least $1/K$. Retain the words stopping at that length. They have $W_X \leq \delta$, and hence $q^k/K \leq T_k \leq |D|\delta^f$, as required. $\square$

Proof of Theorem 3.1. Apply Lemma 3.2 separately in each component with the same K and $\delta = \rho^K$. Obtain lengths $k_i \in [1, K]$ and word sets D_i . Their moduli $b_i^{k_i}$ are coprime perfect squares, and Lemmas 2.5 and 2.6 give ranks below $H = \lceil 1/(\sigma\delta) \rceil$ in every component. Put $M = \prod_i b_i^{k_i}$ and $Q = \prod_i |D_i|$. Lemma 2.4 gives a ranked alphabet of size Q modulo M with height $h = 1 + n(H - 1)$. Multiplying the selected-moment estimates yields

$$M^\alpha \leq Q\delta^F K^n. \tag{3}$$

Since $\delta \leq 1$, the fixed constant $c = (n+1)(1+\sigma^{-1})$ satisfies $h \leq c/\delta$. Consequently

$$(Mh)^\alpha \leq Qc^\alpha K^n \delta^{F-\alpha}.$$

Because $F - \alpha > 0$ and $0 < \rho < 1$, the last factor $c^\alpha K^n \rho^{K(F-\alpha)}$ tends to zero. For example, the ratio of consecutive terms tends to $\rho^{F-\alpha} < 1$, which proves this without any asymptotic counting formula. Choose and fix K so that $(Mh)^\alpha \leq Q$.

Write $L = Mh > 1$. Repeating this one ranked alphabet t times and applying Lemmas 2.5 and 2.3 gives Q^t square-difference-free integers in $[1, L^t]$. For any $N \geq L$, let t be the integer with $L^t \leq N < L^{t+1}$. Then

$$N^\alpha \leq L^{(t+1)\alpha} \leq Q^{t+1}, \quad D(N) \geq Q^t \geq Q^{-1} N^\alpha.$$

For every $\varepsilon > 0$, $N^\varepsilon \geq Q$ eventually, giving $D(N) \geq N^{\alpha-\varepsilon}$ for every sufficiently large N . All alphabets, K , and hence L, Q have been fixed before N varies. $\square$

4 Odd-prime digit constructions

4.1 Prime chains

Lemma 4.1. *Let $p \equiv 3 \pmod{4}$ be prime and let $(s_0, \dots, s_{t-1})$ be distinct residues such that $s_j - s_i$ is a nonzero square modulo p whenever $i < j$. Then the pt digits $s_j + pu$, $0 \leq u < p$, form an ordered interval alphabet modulo p^2 , with interval $[j/t, (j+1)/t]$ on each digit $s_j + pu$. Its moment is pt^{1-f} .*

Proof. The element -1 is not a square modulo p : otherwise an element of order four would exist in the multiplicative group of order $p-1$, contradicting $4 \nmid p-1$. A square difference between different low digits therefore points from the earlier to the later chain entry. If the low digits agree, a nonzero difference has valuation one, and cannot be a square modulo p^2 . The intervals are ordered in the first case and no edge is possible in the second. There are p free copies of each interval, giving the moment. $\square$

This is the local Paley-chain mechanism of [9]; the alternating free-digit principle goes back to [3].

4.2 Separate first differences at different primes

Fix distinct odd primes $p_1, \dots, p_s$ and positive integers $e_1, \dots, e_s$. A low point a consists of words $(a_{p,0}, \dots, a_{p,e_p-1})$ with $0 \leq a_{p,j} < p$. For distinct low points a, b , impose an edge $a \rightsquigarrow b$ if, for every prime p , either the coordinate words are identical or their first differing position j satisfies

$$b_{p,j} - a_{p,j} \in \mathcal{Q}(p) \setminus \{0\}. \quad (4)$$

The first differing position is allowed to depend on p . No comparison is required from a prime whose entire low word is unchanged.

Lemma 4.2 (Prime-coordinate free-digit lift). *Suppose a finite set S of distinct low points has positive intervals in $[0, 1]$ ordered along every edge $\rightsquigarrow$. Put $F = \prod_p p^{e_p}$. For each $a \in S$, choose arbitrary $u_{p,j} \in \{0, \dots, p-1\}$ and form*

$$x_p = \sum_{j=0}^{e_p-1} (a_{p,j} p^{2j} + u_{p,j} p^{2j+1}) \pmod{p^{2e_p}}. \quad (5)$$

Combining the x_p by CRT produces an ordered interval alphabet in base F^2 , with $F|S|$ digits. Every free copy uses its low point's interval, and the moment is $F \sum_{a \in S} w_a^f$.

Proof. Unique prime-adic digit expansions and CRT show that all these digits are distinct and count their number. Suppose different lifted digits x, y have a square difference modulo F^2 . Consider a prime coordinate in which they differ. If $k < 2e_p$ is the first differing actual digit position, then $y_p - x_p$ is divisible by p^k but not by p^{k+1} : all lower digits agree and the coefficient at p^k is nonzero modulo p . The congruence to a square forces k to be even. Indeed, if $z^2 \equiv y_p - x_p \pmod{p^{2e_p}}$, the valuation of z^2 below $2e_p$ must equal k , so $k = 2j$; dividing shows that the leading nonzero difference is a square modulo p .

If the low coordinate words differ, the first actual difference must thus occur at their first differing low digit, and (4) holds. If two distinct lifted digits have the same low point in every coordinate, some first actual difference would instead lie in a free, odd position, a contradiction. Thus their low points are distinct, and each coordinate is either identical as a low word or obeys (4). The assumed low interval order therefore gives the required lifted order. Counting free copies gives the moment formula. $\square$

The first differing position must be tested separately at each prime. For example, the 19-coordinate may first differ in position 0 while the 23-coordinate first differs in position 1. If one coordinate word is unchanged, its zero difference is allowed. The lemma therefore does not require the difference to be invertible modulo the composite base. Only the explicit low points and their interval order enter the proof.

4.3 The two small composite inputs

For the first component use $p = 5, 43$, with one low digit in each coordinate. Table 1 specifies all 17 points and widths. Every width is its displayed numerator divided by 10,000. For the edge

Width numerator	Low points (a_5, a_{43})
1181	$(0, 18), (3, 25), (1, 27), (4, 29)$
1181	$(0, 24), (0, 22), (3, 42), (2, 5)$
579	$(1, 19), (2, 40), (3, 22), (4, 35)$
1476	$(3, 3), (4, 39)$
984	$(0, 33), (1, 28), (2, 20)$

Table 1: The complete 17-point input at 5 and 43.

relation in (4), the graph has 78 directed edges and maximum weighted path $9992/10000 < 1$. Thus Lemma 2.2 provides the intervals. Its low-point moment is

$$Z_{215}(f) = 8(1181/10000)^f + 4(579/10000)^f + 2(1476/10000)^f + 3(984/10000)^f.$$

With $f = 0.0268$, exact arithmetic gives

$$\log Z_{215}(f) - (2\alpha_0 - 1) \log 215 > 0.00008383. \quad (6)$$

By Lemma 4.2, the resulting alphabet in base 215^2 therefore has moment greater than $(215^2)^{\alpha_0}$.

For the second component use $p = 19, 23$, with two low digits at each prime. Its 729 points have a short description. Let

$$S = \{(a_i, b_i) : 0 \leq i < 27\} \subseteq \mathbb{F}_{19} \times \mathbb{F}_{23}$$

be the seed in Table 4. The same table specifies four integers (A_i, B_i, C_i, D_i) for each seed point. Take all ordered pairs (i, j) and set

$$\begin{aligned}(a_{19,0}, a_{19,1}) &= (a_i, A_i a_j + B_i \bmod 19), \\ (a_{23,0}, a_{23,1}) &= (b_i, C_i b_j + D_i \bmod 23).\end{aligned}\tag{7}$$

Thus each first low digit selects an affine copy of the same seed for the second low digit. The multipliers A_i, C_i are nonzero squares in their respective fields. They preserve the square relation inside a fiber. Varying the copies between fibers changes the constraints imposed by pairs whose prime coordinates first differ at different positions. The proof uses the explicit points (7); no general claim that every such choice is feasible is needed.

The accompanying array assigns each point a positive width $n_{ij}/10000$. These widths are chosen jointly, not by multiplying seed widths. The full graph has 109,116 edges; its maximum weighted path is $9972/10000 < 1$. For $f = 0.0673$ the exact moment check gives

$$\log \sum_{i,j=0}^{26} (n_{ij}/10000)^f - 2(2\alpha_0 - 1) \log 437 > 0.00025511.\tag{8}$$

Lemma 4.2 now gives an alphabet in base 437^4 with moment greater than $(437^4)^{\alpha_0}$. The table of 729 width numerators has 165 distinct values and is supplied as machine-readable data rather than as several pages of integers. Section 7 specifies its format and the exact checks that certify both statements.

5 A binary recursion with parity windows

The prime 2 does not furnish an odd-prime chain. Its useful local fact is instead that every odd square is 1 modulo 8. A branch fixes a digit modulo 4, and the parity of its remaining tail determines the digit modulo 8. Thus two windows per state contain exactly the information needed to order square differences between branches.

5.1 States and transitions

A state s has an even window E_s and an odd window O_s in $[0, 1]$. A window may be absent; each present window is a closed interval of positive length, and at least one window is present. At depth m , the state will describe an ordered interval alphabet modulo 4^m , with the interval of each digit inside the window of its parity.

A transition from s has at most one branch for each $r \in \{0, 1, 2, 3\}$. A branch specifies a child j , scale $u > 0$, shift $t \in \mathbb{R}$, and bits $\tau, \eta \in \{0, 1\}$, called swap and reflect. For a child digit x modulo $Q = 4^{m-1}$, let

$$x' = ((-1)^\eta x + \tau) \bmod Q, \quad y = r + 4x'.\tag{9}$$

When $\eta = 1$ reflect the child interval $[a, a + w]$ to $[1 - a - w, 1 - a]$; then apply the increasing affine map $z \mapsto t + uz$. The parity of x' is the parity of x toggled by τ , since Q is even. For transformed parity $p \in \{0, 1\}$, let J_{r+4p} be the placed window obtained from the child window of parity p xor τ , with the same reflection and affine map. An absent child window contributes no J_{r+4p} .

The transition conditions are

$$J_{r+4p} \subseteq E_s \quad (r \text{ even}), \quad J_{r+4p} \subseteq O_s \quad (r \text{ odd}),\tag{10}$$

$$\sup J_k \leq \inf J_{k+1 \bmod 8} \quad \text{whenever both windows are present.}\tag{11}$$

In particular, (11) includes the comparison from class 7 to class 0. Missing classes make such a cyclic specification feasible.

Lemma 5.1 (Binary transition). *For $m \geq 2$, if the child states are valid at depth $m - 1$ and (10)–(11) hold, the branches construct a valid state at depth m .*

Proof. Different branches give different residues modulo four, and within a branch (9) is injective. Thus the digits are distinct. Containment of intervals follows from (10).

Suppose $Y - X$ is a nonzero square modulo 4^m . If the digits belong to one branch, then $Y - X = 4(y' - x')$ modulo 4^m . Any corresponding square root is even, so division by four gives a square difference modulo Q . Without reflection the child order applies directly. With reflection the transformed difference equals the negative of the original child difference; consequently the child square edge has the opposite orientation. Reflection of the intervals reverses that order once more, giving the required transformed interval order. Translation cancels in differences, and the positive scale preserves order.

For different branches whose residues differ by two modulo four, the difference is 2 modulo four and is not a square. In the remaining case the difference is odd. Every odd square is 1 modulo eight, since $(2k+1)^2 = 4k(k+1) + 1$. Therefore the numerical classes of X, Y are $k, k+1$ cyclically modulo eight. Their intervals lie in the corresponding J windows, so (11) orders them. This exhausts all cases. Only the necessary condition on odd squares is used. $\square$

5.2 Moment growth and a finite depth

Fix $0 < f < 1$. Let $v_s > 0$ and $\lambda > 1$ satisfy

$$\sum_{\text{branches } s \rightarrow j} u^f v_j \geq \lambda v_s \quad \text{for every } s. \quad (12)$$

This positive-vector certificate is a familiar device for bounding the growth of a nonnegative matrix [11]. Here the required consequence is just the following elementary induction.

Lemma 5.2. *Let d_s be the length of the wider present window of state s , and put $c = \min_s d_s / v_s > 0$. The transitions construct depth- m states with moments $Z_{s,m} \geq c\lambda^{m-1}v_s$ for all $m \geq 1$.*

Proof. At depth one choose just the digit 0 or 1 of the parity of a widest window (either choice is allowed in a tie) and assign it the whole window. A singleton has no nontrivial square edge. Its moment d_s^f is at least d_s , since $0 < d_s \leq 1$. This gives $Z_{s,1} \geq cv_s$. Reflection and translation do not change widths, scaling multiplies each moment by u^f , and different branches are disjoint. Thus

$$Z_{s,m} = \sum_{s \rightarrow j} u^f Z_{j,m-1} \geq c\lambda^{m-2} \sum_{s \rightarrow j} u^f v_j \geq c\lambda^{m-1}v_s.$$

Validity follows inductively from Lemma 5.1. $\square$

If a root state has both windows $[0, 1]$ and $v_{\text{root}} = 1$, it produces an unrestricted interval alphabet modulo 4^m . Shrinking every final interval at its left endpoint by a factor of $1/2$ preserves all orders, makes all widths at most $1/2$, and multiplies the moment by 2^{-f} . Therefore the condition

$$\log c + (m-1) \log \lambda - f \log 2 - \alpha m \log 4 > 0 \quad (13)$$

implies the required moment $Z_f > (4^m)^\alpha$. This uses only a fixed finite depth, even if its numerical value is enormous.

5.3 The explicit seven-state example

Use states $0, \dots, 6$, with root 1 and windows in Table 2. All 26 transitions are in Table 5. Together these two tables specify the entire recursion, including all shifts and reflections.

State	Even window E_s	Odd window O_s
0	$[0, 1]$	absent
1	$[0, 1]$	$[0, 1]$
2	$[0, .22]$	$[\cdot 51, 1]$
3	$[0, 1]$	$[\cdot 22, \cdot 51]$
4	$[0, \cdot 33]$	$[\cdot 40, 1]$
5	$[0, \cdot 52]$	$[\cdot 22, 1]$
6	$[0, 1]$	$[\cdot 33, \cdot 40]$

Table 2: The seven parity-window states.

The root illustrates how the geometry works. Its nonempty placed windows are

$$\begin{aligned} J_0 &= [0, 49/178], & J_1 &= [49/178, 39/89], & J_2 &= [39/89, 65/89], \\ J_3 &= [65/89, 1], & J_5 &= [0, 50/89], & J_6 &= [50/89, 1]. \end{aligned}$$

Classes 4 and 7 are absent. The required comparisons are precisely $J_0 \leq J_1 \leq J_2 \leq J_3$ and $J_5 \leq J_6$, where an inequality means that the left interval ends before the right one starts. They hold by the displayed endpoints. The other states are checked in the same way. In total there are 86 endpoint containment and order comparisons, all between short rational numbers.

Fix $f_B = 0.154$ and write $\phi(t) = t^{f_B}$. The moment transition matrix, with rows and columns ordered from 0 to 6, is explicitly

$$M = \begin{pmatrix} 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ u & 0 & 0 & \beta & 0 & \beta & 0 \\ v & 0 & 0 & 0 & 0 & 0 & 0 \\ w & 0 & 1 & 0 & 0 & 0 & 1 \\ x & 0 & 0 & 0 & 0 & 0 & 0 \\ y & z & 0 & 0 & 0 & 0 & 0 \\ t & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

Here

$$\begin{aligned} u &= \phi(49/178) + \phi(24/89), & \beta &= \phi(50/89), \\ v &= 2\phi(\cdot 22) + 2\phi(\cdot 49), & w &= 2\phi(\cdot 11), \\ x &= 2\phi(\cdot 33) + 2\phi(\cdot 60), & y &= \phi(\cdot 22) + \phi(\cdot 48), \\ z &= 2\phi(\cdot 15), & t &= 1 + 2\phi(\cdot 07). \end{aligned}$$

To prove growth, we do not need to compute an eigenvalue. Use the exact positive vector

$$V = (.69921, 1, .82524, .94106, .86406, .93399, .87113)^\top \quad (14)$$

and the rational number $\lambda = 2.8602$. Exact bounds for the fractional powers establish, in each of the seven rows,

$$(MV)_s - \lambda V_s > 0.0001195. \quad (15)$$

In the growth lemma the initialization constant is $c = 12250/20631$, attained at state 2. Set $m = 100000$. The two useful numerical inequalities are

$$\log \lambda - \alpha_0 \log 4 > 0.00007904, \quad (16)$$

$$\log c + (m - 1) \log \lambda - f_B \log 2 - \alpha_0 m \log 4 > 6.225. \quad (17)$$

Consequently, halving the final widths produces a valid alphabet in base 4^{100000} with moment greater than $(4^{100000})^{\alpha_0}$. The depth is large but fixed; induction represents the alphabet, so its digits never have to be enumerated. Every resulting width is positive and at most $1/2$.

6 Combining the nine components

The six prime chains and their exact rational powers are in Table 3. For a chain of length t modulo p , Lemma 4.1 gives moment pt^{1-f_p} in base p^2 . The required inequality follows from

$$(1 - f_p) \log t - (2\alpha_0 - 1) \log p > 0.$$

Every row passes this exact comparison; these powers are deliberately rounded down and are not individually optimized. Their sum is 0.51011.

p	t	Ordered chain	f_p
3	2	0, 1	.18215
7	3	0, 1, 2	.08603
11	4	0, 1, 4, 5	.10746
31	7	0, 1, 8, 5, 2, 9, 10	.08940
59	9	0, 1, 28, 17, 22, 4, 26, 20, 29	.04242
103	11	0, 1, 29, 92, 2, 61, 17, 30, 93, 18, 19	.00265

Table 3: Six chains of square differences. All forward differences are nonzero squares modulo p , and all backward differences are nonsquares.

Proof of Theorem 1.1. The component bases and contributions are

square bases	contributions
$3^2, 7^2, 11^2, 31^2, 59^2, 103^2$	the six powers in Table 3
215^2	.0268
437^4	.0673
4^{100000}	.154.

Their underlying prime factors are respectively 3, 7, 11, 31, 59, 103, {5, 43}, {19, 23}, 2, so the bases are pairwise coprime perfect squares. The chain lemma, (6), (8), and (17) establish the required moment inequalities. Their contributions add to 0.75821, leaving the exact surplus

$$\sum_i f_i - \alpha_0 = 0.000209 > 0.$$

Every alphabet is now finite and fixed. The chain widths are less than one, both odd inputs have widths strictly between zero and one, and the binary widths have been halved. Thus common constants $0 < \sigma \leq \rho < 1$ exist. Theorem 3.1 applies and proves the result for every sufficiently large N . $\square$

The fixed binary depth is chosen before taking $N \rightarrow \infty$. Its size affects constants and the threshold for N , but causes no uniformity issue in the interval-moment criterion. In fact that criterion constructs one fixed block whose repetitions give $D(N) \geq cN^{\alpha_0}$ for a fixed $c > 0$ and all sufficiently large N .

7 What the finite certificate checks

The general proof has reduced the theorem to finite statements about explicit integer arrays and rational numbers. This section explains their verification. The attached Python program contains the complete data and checkers, uses only the standard library, and makes no network requests.

Odd geometry. The files `data/q215.json` and `data/q437.json` list the low points and their width numerators. For a list of primes, each point contains one low word for each prime, in increasing digit significance. The common width denominator is 10,000. Thus the latter file specifies all n_{ij} in (8), with the point itself identifying its pair (i, j) . A separate seed-and-map check regenerates (7) and verifies that it agrees with the supplied array.

The checker verifies primality, point distinctness, digit ranges, positive widths strictly below one, and the full edge relation (4) on every ordered pair of distinct points. All edges must point forward in the supplied ordering. For width numerator n_j , compute

$$d_j = n_j + \max(\{d_i : i \rightsquigarrow j\} \cup \{0\}).$$

Induction in that ordering shows that d_j is the maximum path numerator ending at j . The check $\max_j d_j \leq 10000$ therefore proves feasibility by Lemma 2.2, and also defines the interval starts as $(d_j - n_j)/10000$. This checks all 78 and 109,116 edges; no graph sampling or assumption about a transitive reduction is involved.

Binary geometry and prime chains. The file `data/binary.json` records the windows, branches, vector, growth constant, moment power and depth displayed above. The checker constructs every placed window J_k with exact fractions and checks all conditions (10)–(11). It also checks the valid state indices, unique branch residues, positive scales, and the two bits. These 86 comparisons, together with the binary lemma, certify every depth. A direct check at depth 5 is an additional sanity check and is not used in place of induction. The six chains are checked in both orientations on all 244 ordered pairs of distinct entries. Primality and disjointness of all CRT prime factors are checked as well.

Exact arithmetic for the moments. For completeness we give the elementary enclosure used to check fractional powers. If $1 \leq x \leq 2$ and $z = (x - 1)/(x + 1)$, then $0 \leq z \leq 1/3$ and

$$L_n(x) := 2 \sum_{j=0}^{n-1} \frac{z^{2j+1}}{2j+1} \leq \log x \leq L_n(x) + \frac{2z^{2n+1}}{(2n+1)(1-z^2)}. \quad (18)$$

Indeed, integrate the geometric series to expand $\log((1+z)/(1-z))$. Bound the tail by replacing its denominators by $2n+1$ and summing a geometric series. For an arbitrary positive rational x , write $x = 2^k y$, $1 \leq y \leq 2$, and bound $k \log 2 + \log y$, reversing endpoints when multiplying by a negative k . All these bounds are rational.

Write $\ell(x), u(x)$ for the resulting lower and upper bounds on $\log x$. A proposed positive rational lower bound L for x^f , $f \geq 0$, is accepted only if

$$u(L) \leq f\ell(x).$$

This proves $L \leq x^f$ by monotonicity of the logarithm. The odd checker uses 32 terms of (18) and the binary checker uses 70. Decimal arithmetic proposes enclosures; exact rational comparisons accept or reject them. Thus the validity of a successful check does not depend on the accuracy of those proposals.

Summing accepted lower bounds for the odd powers and using upper bounds for the subtracted logarithms proves (6) and (8). The same procedure proves every row of (15) and the fixed-depth inequality (17). Each chain power is checked directly as a rational, and the final contribution surplus is exact rational addition. No spectral-radius approximation or asymptotic estimate enters these acceptance tests.

Scope. The computer-assisted step is the execution of these finite integer and rational checks. The mathematical reasons that their acceptance implies the theorem are proved in the preceding sections. The certificate does not assert that the inputs are optimal. Its complete source and data can be extracted and inspected; no missing research notes or search programs are premises of the argument.

A Explicit local tables

Table 4 supplies all seed points and conditional maps in (7). Each row i specifies the first low point (a_i, b_i) and the map $(a, b) \mapsto (A_i a + B_i, C_i b + D_i)$ in $\mathbb{F}_{19} \times \mathbb{F}_{23}$ for its second low point. Table 5 uses the branch convention (9). A blank branch is absent; every listed decimal is exact.

i	a_i	b_i	A_i	B_i	C_i	D_i
0	10	6	1	0	1	0
1	14	6	1	0	12	7
2	2	6	1	0	13	11
3	10	19	1	0	1	0
4	0	16	1	0	1	0
5	0	19	7	6	12	8
6	11	16	1	0	13	10
7	0	20	7	6	1	0
8	11	19	1	0	1	0
9	11	20	1	0	2	10
10	10	9	1	0	1	0
11	16	10	1	0	1	0
12	16	19	1	0	16	17
13	10	22	1	0	1	0
14	17	10	1	0	2	10
15	16	22	1	0	8	20
16	17	19	1	0	2	7
17	14	9	1	9	4	19
18	17	22	5	1	3	17
19	14	22	1	9	1	0
20	2	9	5	2	8	1
21	2	22	1	0	6	20
22	16	12	1	0	1	0
23	0	5	1	0	1	0
24	1	12	1	0	16	11
25	11	5	1	0	13	10
26	17	12	1	0	1	0

Table 4: The 27-point seed and its conditional affine maps for the 729-point component.

Parent s	r	Child j	Scale u	Shift t	Swap τ	Reflect η
0	0	1	1	0	0	0
0	2	1	1	0	0	0
1	0	0	$\frac{49}{178}$	0	0	0
1	1	3	$\frac{50}{89}$	0	1	1
1	2	5	$\frac{50}{89}$	$\frac{39}{89}$	0	0
1	3	0	$\frac{24}{89}$	$\frac{65}{89}$	0	0
2	0	0	0.22	0	0	0
2	1	0	0.49	0.51	1	0
2	2	0	0.22	0	0	0
2	3	0	0.49	0.51	0	0
3	0	2	1	0	0	0
3	1	0	0.11	0.22	0	0
3	2	6	1	0	1	0
3	3	0	0.11	0.4	0	0
4	0	0	0.33	0	0	0
4	1	0	0.6	0.4	1	0
4	2	0	0.33	0	0	0
4	3	0	0.6	0.4	0	0
5	0	0	0.22	0	0	0
5	1	1	0.15	0.22	0	0
5	2	1	0.15	0.37	0	0
5	3	0	0.48	0.52	0	0
6	0	4	1	0	0	0
6	1	0	0.07	0.33	0	0
6	2	0	1	0	1	0
6	3	0	0.07	0.33	0	0

Table 5: All 26 branches of the seven-state recursion. Each row means $x' = ((-1)^\eta x + \tau) \bmod 4^{m-1}$, $y = r + 4x'$, with the interval reflected if $\eta = 1$ and then mapped by $z \mapsto t + uz$.

B Reproducing the checks

The L^AT_EX source contains the complete standalone verifier. Compilation writes it to a separate Python file and attaches the program to the PDF:



Complete certificate and verifier

Save the attachment, or use the separate copy supplied with the paper, and run

```
python3 sarkozy-simple-certificate.py
```

To retain the unpacked source, JSON inputs and reports in a fresh directory, run

```
python3 sarkozy-simple-certificate.py --extract certificate-check
```

The program rejects Python’s `-O` and `-OO` modes, so all verification assertions execute. It checks the identities of the bundled inputs as well as their mathematical conditions. It needs standard Python 3 and no numerical optimization packages.

To compile from the single L^AT_EX source, use an ordinary TeX Live installation and run the following command until cross-references settle (three runs suffice):

```
pdflatex -interaction=nonstopmode -halt-on-error \  
-no-shell-escape sarkozy_simple_0.758.tex
```

The bibliography and certificate are embedded in that source; no external bibliography or certificate download is needed. The tables and explicit parameters above are part of the mathematical specification, and the attachment provides the remaining width array and executable checks.

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